CS 301 — Discrete Mathematics

Graph Theory: Planarity and Coloring

Lecture 12

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1 Introduction to Graph Theory

Definition 1 *A **graph** $G = (V, E)$ consists of a finite set V of vertices and a set $E \subseteq V \times V$ of edges.*

Example. Consider the graph $G = (\{1, 2, 3, 4\}, \{(1, 2), (2, 3), (3, 4), (4, 1)\})$. This is a simple cycle on four vertices, denoted C_4.

Theorem 2 (Handshaking Lemma) *For any undirected graph $G = (V, E)$:*

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. Each edge contributes exactly 1 to the degree of each of its two endpoints, so the total sum of degrees counts every edge twice.

□

2 Trees and Connectivity

Definition 3 *A **tree** is a connected acyclic graph. A **forest** is a disjoint union of trees.*

Lemma 4 *Every tree with n vertices has exactly $n - 1$ edges.*

Proof. By induction on n. The base case $n = 1$ is trivial. For the inductive step, remove a leaf vertex (which always exists in a tree with $n \geq 2$). The resulting graph is a tree on $n - 1$ vertices with $(n - 1) - 1 = n - 2$ edges by the inductive hypothesis. Adding back the leaf restores one edge, giving $n - 1$.

□

Corollary 5 *A connected graph with n vertices and $n - 1$ edges is a tree.*

Theorem 6 (Euler's Formula for Planar Graphs) *For any connected planar graph with V vertices, E edges, and F faces:*

$$V - E + F = 2.$$

Proof sketch. Start with a spanning tree ($E = V - 1$, $F = 1$). Each additional edge adds exactly one face. By induction, $V - (V - 1 + k) + (1 + k) = 2$.

□

3 Worked Examples

Example. Determine whether the graph $K_{3,3}$ (the complete bipartite graph with three vertices in each partition) is planar.

Solution. $K_{3,3}$ has $V = 6$ vertices and $E = 9$ edges. If planar, by Euler's formula: $F = 2-V+E = 2-6+9 = 5$. But every face in a bipartite graph has at least 4 edges, so $2E \geq 4F \implies 18 \geq 20$, a contradiction. Hence $K_{3,3}$ is **not** planar.

Remark. The same technique shows that K_5 is not planar: $V = 5$, $E = 10$, $F = 7$, and $2E = 20 \geq 3F = 21$, contradiction.